Sneha Gathani

4th year PhD Student in Computer Science

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RESEARCH INTERESTS

Data-Driven Decision-Making Systems, Interactive Visual Data Systems, Human-Centered AI

EDUCATION

PhD in Computer Science, *University of Maryland, College Park, GPA: 3.77/4.* — *College Park, US*
Advisor: *Leo Liu* — *June 2020 – May 2025 (expected)*

Masters in Computer Science, *University of Maryland, College Park, GPA: 3.77/4.* — *College Park, USA*
Advisor: *Leilani Battle* — *Aug 2018 – May 2020*

PUBLICATIONS

EuroVis 2022 Full paper | Talk	P.4	**A Grammar-Based Approach for Applying Visualization Taxonomies to Interaction Logs** **Sneha Gathani,** Shayan Monadjemi, Alvitta Ottley, Leilani Battle
CIDR 2022 Full paper | Talk	P.3	**Augmenting Decision Making via Interactive What-If Analysis** **Sneha Gathani,** Madelon Hulsebos, James Gale, Peter J. Haas, Çağatay Demiralp
CIDR 2022 1-page abstract	P.2	**Making Table Understanding Work in Practice** Madelon Hulsebos, **Sneha Gathani,** James Gale, Isil Dillig, Paul Groth, Çağatay Demiralp
CHI 2020 Full paper | Talk	P.1	**Debugging Database Queries: A Survey of Tools, Techniques, and Users** **Sneha Gathani**, Peter Lim, Leilani Battle

WORK EXPERIENCE

Salesforce/Tableau Research — *Seattle, USA*
Research Intern, Mentor: *Arjun Srinivasan* — *Summer 2023*

InsightCurator: An Interactive Curation Interface of Automated Data Insights
Developed an interactive system that allows users to **edit and customize automatically generated data insights** through direct manipulation of chart elements and a manual specification interface. **Target:** *VIS 2024*

Sigma Computing Inc. — *San Francisco, USA*
Research Intern, Mentor: *Çağatay Demiralp* — *Summer, Fall 2021*

Augmenting Decision Making via Interactive What-If Analysis
Delivered Decision Studio, an interactive visual data analysis system that enables business users to **understand input data drivers-output KPI metric relationships** through what-if scenarios using four predictive and prescriptive (PPA) functionalities; validated through three common business uses with Sigma employees. **Paper:** P.3 ***(CIDR 2022)***

RESEARCH

PhD Graduate Research — *UMD, College Park, USA*

An Algebra of What-If Analysis — *Fall 2022 – Present*
Formulating a standardized algebra for data-driven what-if analysis and implementing it is a declarative specification language to enable the scaling of data-driven decision-making processes. **Target:** *VIS 2024*

Understanding Business Users' Data-Driven Decision-Making: Practices, Challenges, and Opportunities — *Fall 2021 – 2023*
Conducted an interview study with professional business users (i.e., sales, marketing, product, and operations managers) to contribute a model of their data-driven decision-making processes elaborating their tasks, tools, and challenges. Performed a follow-up task-based study to understand the role and potential of advanced predictive and prescriptive analytics (PPA) tools in their business decision making. **Under Review:** *TVCG 2024*

Masters Graduate Research — *UMD, College Park, USA*

A Grammar-Based Approach for Applying Visualization Taxonomies to Interaction Logs — *Summer 2020 – Fall 2021*
Translated and demonstrated the applicability of theoretical visualization task **taxonomies** on empirical interaction logs via developing **programmatic mappings** of taxonomies. **Paper:** P.4 ***(EuroVis 2022)***

TraceInspector: A Visualization-based Reverse Engineering Tool for Android Apps — *Spring 2019 – 2020*
Developed TraceInspector, an interactive visualization tool that helps novice analysts **reverse engineer Android apps** for potential security and privacy vulnerabilities.

Debugging Database Queries: A Survey of Tools, Techniques and Users — *Summer 2019*
Performed interdisciplinary literature review and conducted interview study to **understand database query debugging strategies and tools**. **Paper:** P.1 ***(CHI 2020)***